



Analysis of Algorithms
CS 312 - HBD1
Department of Physics and Computer Science
Medgar Evers College
Exam 1 - Redo

Instructions:

- The exam requires completing a set of tasks by the end of the day of 10/08.
- Submission must be uploaded to the Exam01 directory of your GitHub repository.
- Submit each task as a separate .txt or .cpp files.
- Notes are not allowed.
- Cheating of any kind is prohibited and will not be tolerated.
- Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the exam.

TO ACKNOWLEDGE THAT YOU HAVE READ AND UNDERSTOOD THE INSTRUCTIONS ABOVE,
PRINT YOUR NAME AND THE DATE ON YOUR SUBMISSIONS

Grading

Section	Maximum Points	Points Earned
1	2	
2	1	
3	3	
4	2	
5	2	
Total	10	

1. Find a closed-form solution of:

$$\sum_{i=1}^{50} (6i^2 - 4i + 5)$$

2. List the following big-oh runtimes in descending order of growth rate (from fastest-growing to slowest-growing):

$O(n)$	$O(\lceil \lg^2(n) \rceil)$	$O(\lceil \log(n) \rceil)$	$O(\lceil \sqrt{n} \lg(n) \rceil)$	$O(\lceil \sqrt{n^3} \rceil)$
--------	-----------------------------	----------------------------	------------------------------------	-------------------------------

where $\lg^2(n) = (\log_2(n))^2$ and $\log(n) = \log_{10}(n)$.

3. Calculate the runtime function for the worst-case scenario of the following algorithm by constructing its runtime table. Begin by defining the input size, and assign a cost of 1 to every operation. Assume the input array A contains only distinct values.

```

Algorithm(A,B)
01  1. i ← 0
02  2. j ← 0
04  3. s ← 0
05  4. while i < A.length and j < B.length, do
06      1. if A[i] and B[j], then
07          1. s ← s + 1
08          2. i ← i + 1
09          3. j ← j + 1
10  5. while i < A.length, do
11      1. if A[i], then
12          1. s ← s + 1
13          2. i ← i + 1
14  6. while j < B.length, do
15      1. if not B[j], then
16          1. s ← s - 1
17          2. j ← j + 1
18  7. return s

```

4. Write pseudocode for an algorithm named C() that takes an integer parameter n and returns $\sum_{i=1}^n i^3$ if n is positive, and 0 otherwise. The worst-case running time of your algorithm must be $O(1)$ (constant). Prove that your algorithm is correct.
5. Prove each of the following asymptotic notation boundaries directly from their definitions and the rules of limits.
 - a) $2n^2 + 8n - 10 = o(n^3)$
 - b) If $f(n) = \Theta(g(n))$, then $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.